ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B.Tech.(CSE) (AI)

Semester: IV

SUBJECT CODE: BCSAI404

SOFTWARE ENGINEERING

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. **(10 Marks)**

Q. 1. Answer the following questions: **(1 × 5 = 5 Marks)**

(i) Name the primary components of a use case diagram.

(ii) What is the key difference between cohesion and coupling?

(iii) What does functional cohesion mean?

(iv) What are the different components of software?

(v) What is the purpose of a data flow in a DFD?

Q. 2. Answer the following questions: **(1 × 5 = 5 Marks)**

(i) What type of diagram is used to visually represent the interactions between actors and the system?

(ii) How many stages does a SDLC have name all of them?

(iii) What is system testing?

(iv) What is cohesion?

(v) Define Alpha testing and Software reliability.

SECTION – B

NOTE:Attempt all questions. **(5×2×2 = 20 Marks)**

Q.3.(i)(a) Explain the characteristics of software and its importance in software engineering.

OR

(b) Compare and contrast the Waterfall Model and Spiral Model in software development.

(ii)(a) Describe the principles of software engineering and their significance in developing reliable software.

OR

(b) What are Requirements Elicitation Techniques? Explain any two techniques in detail.

Q.4.(i)(a) Compare functional testing with structural testing, providing examples for each.

OR

(b) Discuss the importance of test case specification in the software testing process.

(ii)(a) What are Alpha and Beta testing? Explain their roles in software development.

OR

(b) Describe the COCOMO model for cost estimation, including its types.

SECTION – C

NOTE: Attempt all questions. **(10 × 2 = 20 Marks)**

Q.5.(a) Create a use case diagram for an online shopping system, including actors such as customers, administrators, and payment systems. Specify the relationships between the actors and use cases.

OR

(b) Discuss ISO 9000 certification and its importance for quality assurance in the software industry.

Q.6.(a) What are size metrics like LOC, Token Count, and Function Count? And software project is estimated to have a size of 1000 lines of code (LOC). Using the Basic COCOMO model, calculate the estimated effort in person-months. (Value of a & b in cost estimation is 2.4 and 1.05 respectively).

OR

(b) Define a Data Flow Diagram (DFD) and explain its key components: processes, data stores, data flows, and external entities. Draw a DFD for inventory management system.